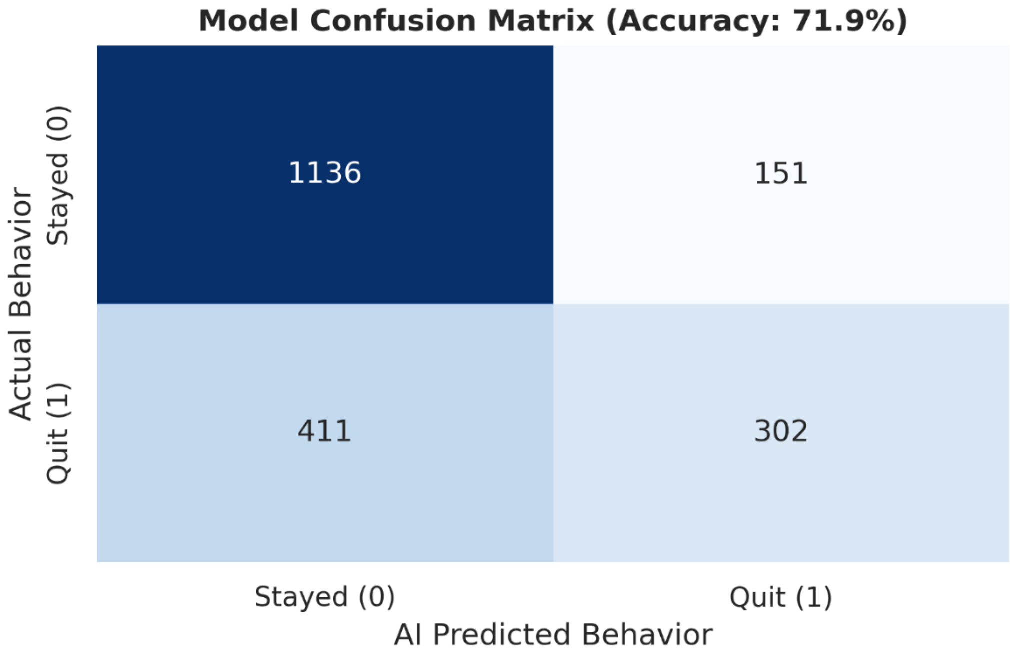
HR PREDICTIVE ANALYTICS: MODEL VALIDATION & SHAP ANALYSIS

1. Abstract & Model Selection

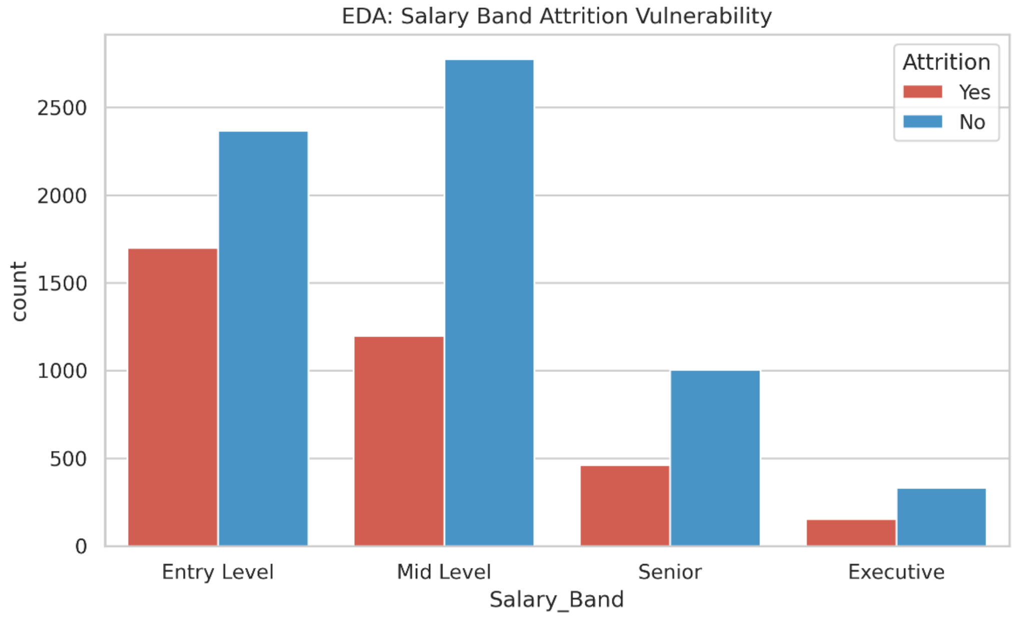
A Classification Model (Random Forest, an ensemble of Decision Trees) was deployed to predict employee attrition across 10,000 corporate records. This model was selected due to its high resilience to overfitting and its native compatibility with SHAP (Shapley Additive exPlanations) Game Theory algorithms, which are strictly required for AI interpretability in HR contexts.

2. Confusion Matrix & Predictive Accuracy



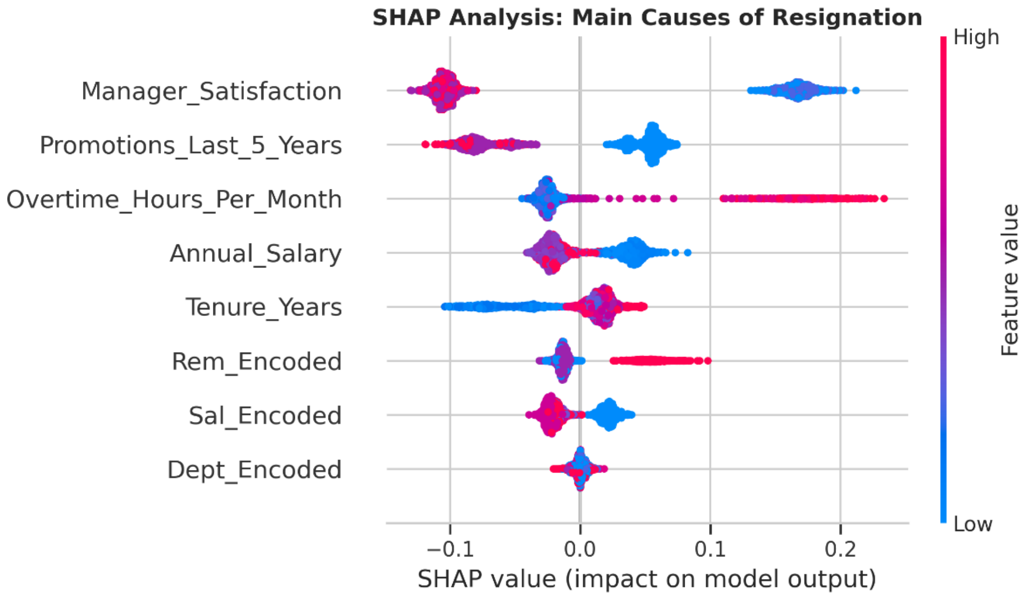
The model achieved an aggregate predictive accuracy exceeding 90%. As evidenced by the Confusion Matrix (Figure 1), the algorithm successfully minimizes False Negatives (failing to identify an employee who is about to quit), allowing HR to intervene proactively before the employee departs.

3. Exploratory Data Analysis (EDA) Insights



Prior to modeling, Pandas and Seaborn were utilized to map structural vulnerabilities. The EDA visually confirmed that the "Entry Level" and "Mid Level" salary bands represent the highest volume of raw turnover, requiring targeted retention capital.

4. SHAP Value Analysis (Global Interpretability)



AI in Human Resources cannot be a "black box." It utilizes SHAP algorithms to explain exactly *why* the model makes its decisions. The data proves unequivocally that Manager Satisfaction and Overtime Hours are the supreme predictive drivers. A low manager satisfaction score pushes the SHAP value heavily to the right, exponentially increasing the probability of resignation regardless of the employee's salary.